

NavalForge Concept

Preliminary conceptual design report

7 m service launch — NF-DEMO-SERVICE-7M — Revision P1

Engineering gate: CONDITIONAL — preliminary results with active engineering reservations

Technical notice: Preliminary engineering result. It does not replace the responsible engineer, formal approval, validated software, detailed structural analysis or verified input data.

Main indicators

Indicator	Value	Unit
Adherence	100.0	%
Displacement	2675.47	kg
Maximum speed	34.00	kn
Required installed power	107.37	kW
Range	128.70	nmi
Corrected GM	0.479	m
Freeboard	0.747	m

Requirement matrix

ID	Requirement	Actual	Result
REQ7-LOA	Overall length shall not exceed 7.50 m	7.0 m	PASS
REQ7-SPEED	Maximum speed shall be at least 30 kn	34.0 kn	PASS
REQ7-RANGE	Range shall be at least 75 nmi	128.69700377961584 nmi	PASS
REQ7-GM	Preliminary corrected GM shall exceed 0.35 m	0.47893410316254525 m	PASS
REQ7-FB	Preliminary freeboard shall exceed 0.40 m	0.7469318304744539 m	PASS
REQ7-PAYLOAD	Payload shall be at least 300 kg	300.0 kg	PASS

Warnings and assumptions

- GZ curve uses a wall-sided preliminary estimate; validate with inclined geometry before statutory use.
- Partially filled tanks present; free-surface correction applied.
- Consumption must be replaced by verified engine/propulsor curves for contractual use.
- All project and price data are synthetic demonstrations.
- Departure condition includes six people and mission payload.
- Conceptual parametric hull used unless verified offsets are supplied.
- Demonstrative prices are synthetic and are not current quotations.

Selected alternatives

NF-ECO — NF-DEMO-SERVICE-7M-V002

Lowest demonstrative cost among variants passing every mandatory requirement.

NF-BALANCED — NF-DEMO-SERVICE-7M-V005

Highest auditable multicriteria balance of cost, performance, range, GM and risk.

NF-PERFORMANCE — NF-DEMO-SERVICE-7M-V007

Highest combined performance and technical-margin screening among compliant variants.